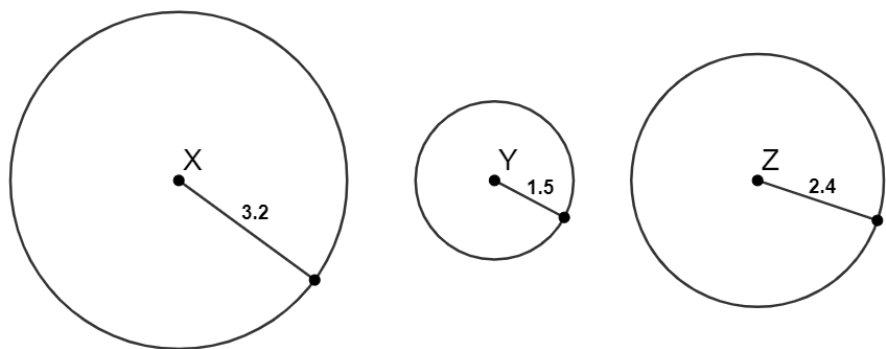


Circles X , Y , and Z are shown. The radius of Circle X is 3.2 units long. The radius of Circle Y is 1.5 units long. The radius of Circle Z is 2.4 units long.



1. Complete the following table. Round to the nearest thousandth. Use the π button on your calculator.

Circle	Radius	Circumference
X	3.2	20.106
Y	1.5	9.425
Z	2.4	15.080

2. Complete the table.

Circle X and Y	
Ratio of Radius of Circle X to Radius of Circle Y	$\frac{3.2}{1.5} = 2.133$
Ratio of Circumference of Circle X to Circumference of Circle Y	$\frac{20.106}{9.425} = 2.133$

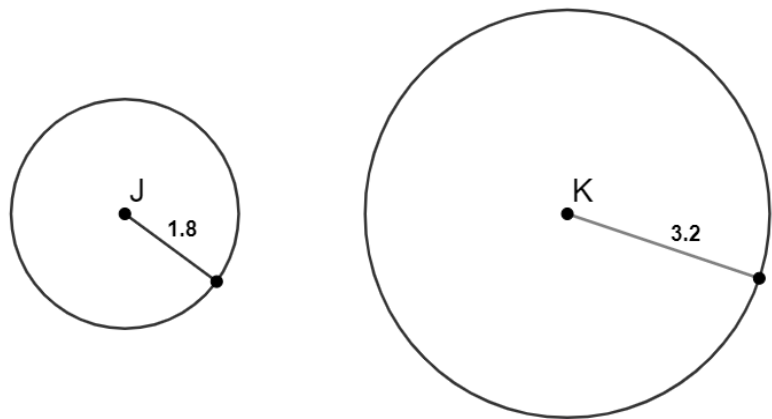
3. Complete the table.

Circle Y and Z	
Ratio of Radius of Circle Y and Radius of Circle Z	$\frac{1.5}{2.4} = 0.625 = \frac{5}{8}$
Ratio of Circumference of Circle Y and Circumference of Circle Z	$\frac{9.425}{15.080} = 0.625 = \frac{5}{8}$

4. Complete the table.

Circle X and Z	
Ratio of Radius of Circle X to Radius of Circle Z	$\frac{3.2}{2.4} = 1.333$
Ratio of Circumference of Circle X to Circumference of Circle Z	$\frac{20.106}{15.080} = 1.333$

Circles *J* and *K* are shown. The radius of Circle *J* is 1.8 units long. The radius of Circle *K* is 3.2 units long.



Show Circle *J* is similar to Circle *K* by completing the proof by contradiction:

- 5. Assume that Circle *J* is **not similar** to Circle *K*.
- 6. The ratio of the radius of Circle *J* to the radius of Circle *K* is **0.563**.
- 7. The ratio of the circumference of Circle *J* to the circumference of Circle *K* is **0.563**.
- 8. The ratio of the radii and the ratio of the circumferences are **equal**, so corresponding parts of Circle *J* and Circle *K* are proportional.
- 9. Since the corresponding parts of Circle *J* and Circle *K* are proportional, the circles are **similar**. This is a contradiction, so the assumption that Circle *J* is not similar to Circle *K* is not true.

10.

✖ All

Some

No

 circles are similar.